Applies to MASD Engineering Toolbox 00.01.000 · Battery Calculator 02.00.002 · Commissioning Report 02.00.001 · MASD engine 1.0.0

MASD Engineering Toolbox — Quick Start

From PipeCAD design to approved as-built: configure, size batteries, hand off, log site changes, verify and approve.

1 · PipeCAD designer

Name Floors as Areas/Groups. Keep detector names ≤16 characters including spaces (display limit). Set display type (Standard/Minimum/Command = partNumber 0/1/2). Place PSU and Battery objects where they will be mounted. Set aspirator speed and transit-time class (60/90/120 s = VEWFD/EWFD/SFD). Run the calculation, then export the .pl — addresses export as 0, that is expected.

2 · Toolbox — Config

Pick the mode (Config, Calculator or Both; saved per project). Import .pl. Each detector and display gets Alert/Prealarm/Fire 1/Fire 2, delays, latching, flow-fault limits and delay, day/night, LDD, I/O assignments and display settings, seeded from the performance class. Badges: **manual** = stated in the ModuLaser Install Manual (flow ±20% and 120 s flow-fault delay, Table 29 p.76); **typical – unverified** = must be checked; **design** = set by you. Then set **Performance Criteria**: UL market (US default) states UL 268, 7th Edition and applies VEWFD 1.0 / EWFD 1.5 / SFD-OAP 2.18 %obs/ft (strictly less than); any other basis needs each value entered, referenced and confirmed. Confirm the auto-drafted narrative and enter the Engineer of Record and spec reference — Export project is blocked until done.

3 · Toolbox — Battery calc

Same engine as the Battery Calculator (MASD engine 1.0.0). Set battery factors, assign PSUs (≤4 detectors + 1 display, 65 Ah limit, never shared across Areas/Groups), set SenseNET addresses. File → Export project writes the .mlproj.json. Old calculator files and Excel exports: File → Import legacy calc / Excel… (migration report shows anything that did not map).

4 · Commissioning Report

Import project: devices, PSUs, batteries, addresses, pipe data and your configuration values arrive prefilled (typical values stay flagged). Every site change to a design value is logged — field, old/new, who, when, why, photo. Technician signs; print the Field Change / As-Built report; export the project back.

5 · Toolbox — Verify & close

Import the returned project, open Field changes: Verify & apply or Reject each change; the battery calc re-runs. Print the PipeCAD change list, update PipeCAD, import the new .pl — matching changes verify automatically. QA/Auditor approves; name, date and status are stored in the file.

Do

- Name Floors as Areas/Groups; keep names ≤16 characters
- Verify every *typical – unverified* value before handoff
- Send the .mlproj.json and the .pl together
- Re-import the updated .pl after PipeCAD changes

Don't

- Share a PSU across Areas/Groups
- Exceed 4 detectors + 1 display or 65 Ah per PSU
- Edit addresses in PipeCAD (set them in the Toolbox)
- Approve as QA with PipeCAD changes still pending .pl

Checklist before sign-off

☐ Mode chosen for the project
☐ Every device configured, no red errors
☐ Typical values verified
☐ PSUs assigned, addresses set, all PASS
☐ .mlproj.json exported with the .pl
☐ Field changes verified / rejected
☐ PipeCAD change list closed by new .pl
☐ QA approval recorded

PipeCAD = layout, pipes, names **Toolbox** = configuration, PSUs, batteries, addresses, change approval **Report** = site results and field changes

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